

Xiaowei Chen

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Academic Appointments

Assistant Professor of Physics Engineering (Tenure-Track) Department of Physics, Astronomy, and Engineering University of Central Arkansas (UCA)	01/2025 - Present Conway, AR, USA
Instructor (Tenure-Track) Department of Physics, Astronomy, and Engineering University of Central Arkansas (UCA)	08/2024 - 12/2024 Conway, AR, USA

Education

Oklahoma State University (OSU) Ph.D. in Electrical & Computer Engineering (GPA: 4.0/4.0)	12/2024 Stillwater, OK, USA
Xi'an Jiaotong University (XJTU) B.E. in Computer Science & Technology (GPA:3.4/4.0)	06/2009 Xi'an, Shaanxi, China

Teaching

Instructor, UCA - Microelectronics - Fall 2024 (4 students), Fall 2025 (17), Fall 2026 (13) - Microcontrollers - Spring 2025 (9 students), Spring 2026 (16) - Singles and Systems - Spring 2025 (3 students), Spring 2026 (3) - Computer Vision - Fall 2025 (8 students), Fall 2026 (9) - College Physics 2 - Summer 2025 (19 students), Summer 2026 (43)	
Teaching Assistant, OSU - Signal Analysis - Spring 2021 (35 students), Spring 2022 (37), Spring 2023 (50), Fall 2023 (50) - Fundamentals of Electrical Circuits - Fall 2021 (71 students), Fall 2022 (72)	

Mentoring

Mentor, UCA, Robotic Vision and Intelligent Systems Lab, 2024 – Present - Mentored undergraduate students in robotics, embedded systems, computer vision, indoor localization, and assistive technologies. Mentored undergraduate students in the development of an intelligent assistive robot integrating embedded systems, computer vision, indoor localization, and mobile robotics. - Supervised research and engineering design projects from concept development through implementation, testing, and presentation. - Guided students in technical writing, research presentations, conference submissions, and preparation for graduate study.	
Mentor, OSU, Visual Computing and Image Processing Lab, 2020 - 2024 Mentored undergraduate and graduate students in image processing and computer vision research.	

Research Funding

UCA Foundation Faculty Grant, University of Central Arkansas Foundation - Title: Developing an Augmented Reality Indoor Assistive Navigator for Smart Campus Environments - Role: Co-PI - Award Period: 2026 - Award Amount: \$1,200	
Faculty Start-Up Research Support, University of Central Arkansas - Role: PI - Award Period: 2024 - 2026 - Award Amount: \$25,000	

Industry Experience

Senior Software Testing Engineer, C.C.H International Group Inc., Zhengzhou, China Led testing and validation of travel websites and statistical systems, including requirements analysis, test design, performance testing, and issue resolution.	2015 – 2017
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Software Testing Engineer, ZTEICT Technology Co., Ltd., *Shenzhen, China*

2012 – 2014

- Tested healthcare software systems, including web applications, mobile applications, and APIs, and developed related testing and technical documentation.

Website Testing Manager, Linekong Technology Co., Ltd., *Beijing, China*

2009 – 2012

- Supervised software testing projects and team activities, including test planning, environment setup, system validation, and defect management.

Technical Skills

- **Programming:** Python, MATLAB, SQL, Java
- **AI & Computer Vision:** PyTorch, TensorFlow, OpenCV, Keras, scikit-image; image classification, object detection, segmentation, pose estimation.
- **Robotics & Embedded Systems:** ROS 2, Raspberry Pi/Pico, embedded sensors and actuators, system integration
- **Optimization & Data Analysis:** Linear Programming, Nonlinear Optimization, Data Augmentation.

Publications

Peer-Reviewed Papers

- **Chen, X.** and Fan, G., 2025. Probabilistic perspective-n-lines for indoor camera pose estimation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW) (pp. 4650-4659).
- **Chen, X.**, Fan, G., Roberts, E., and Howell, S.J. 2025. An Intelligent Assistive Technology to Support In-Home Activities of Daily Living for People with Mild Dementia. Journal of Advances in Information Technology.
- **Chen, X.**, Fan, G., Roberts, E., and Howell, S.J. 2024. A ChatGPT-enabled intelligent assistive technology for homecare to support activities of daily living for people with mild dementia. In Proceedings of the 2024 4th International Conference on Human-Machine Interaction.
- **Chen, X.**, Fan, G., Roberts, E., and Howell, S.J. A Transfer Learning-based Smart Homecare Assistive Technology to Support Activities of Daily Living for People with Mild Dementia. 2023 IEEE 23rd International Conference on Bioinformatics and Bioengineering (BIBE)
- Roberts, E., Fan, G. and **Chen, X.**, 2023. In-Lab Development of a Mobile Interface for Cognitive Assistive Technology to Support Instrumental Activities of Daily Living in Dementia Homecare. Journal of Aging and Environment, pp.127-141.
- **Chen, X.** and Fan, G., 2023. Indoor Camera Pose Estimation from Room Layouts and Image Outer Corners. IEEE Transactions on Multimedia.
- **Chen, X.** and Fan, G., 2022. Egocentric Indoor Localization from Coplanar Two-Line Room Layouts. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW) (pp. 1549-1559).
- **Chen, X.** and Fan, G., 2021. Egocentric Indoor Localization from Room Layouts and Image Outer Corners. In Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW) (pp. 3456-3465).

Peer-Reviewed Abstracts

- Roberts, E., Fan, G. and **Chen, X.**, 2024. In-Lab Testing of Assistive Technology for Dementia Homecare: Sequencing and Delivery Cues for Tasks and Navigation. In EDRA 55 Human-Centric Environments.
- Roberts, E., Fan, G. and **Chen, X.**, 2022. Development of Assistive Technology for Instrumental Activities of Daily Living for Dementia Homecare Environments. Alzheimer's & Dementia, 18, e068099.

Manuscript Submitted

- **Chen, X.**, Ajiatas-Ralac, C., Johnson, Z., Phillips, J., Haring, D. Developing Engineering Readiness Through a Closed-Loop Human-Centered Design Framework in an Undergraduate Microcontroller Course. **Accepted** by the 2026 ASEE Midwest Section Conference.

Manuscript in Preparation

- Chen, X. and Fan, G. End-to-End Probabilistic Perspective-n-Line based on Room Layouts for Camera Pose Estimation.

Selected Honors & Awards

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| • Robberson Research and Creative Activity Grant , OSU, 2023 | • NSF Student Travel Award , IEEE BHI, 2023 |
| • Graduate 3MT (Three Minute Thesis) Competition Finalist , OSU, 2022 | • OSU Research FNDN Chair , OSU, 2022 |
| • Dr Yarlagadda Grad Fellowship , OSU, 2021 | • Leo J. Peters & Josie Mosely Peters Award , OSU, 2020 |

Community Engagement

- **Internal Committee:** Department Chair Search Committee Member, Foundation Scholarship Committee Member
- **Volunteer:** VEX Robotics Competition, Arkansas State Science Fair, UCA STEM Day
- **Journal Reviewer:** IEEE Transactions on Multimedia, Multimedia Tools and Applications
- **Conference Reviewer:** IEEE International Conference on Multimedia and Expo, 2026 ASEE Midwest Section Conference